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Solar Panel Trickle Charging: A Sustainable Solution for Enhancing Battery Lifecycle and Reliability in Cementing Bulk Equipment

Mohd Aliff Che Hazamuddin, M. Shafeek Parattu Veetil, and M. Alhussain, TAQA Well Solutions, Dammam Second Industrial City, Kingdom of Saudi Arabia

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Abstract

The issue of high battery consumption caused by self-discharge presents a significant operational challenge for cementing bulkers, which often remain idle at field locations for extended periods. This paper examines the application of trickle charging systems as a sustainable and maintenance-free method for maintaining battery charge during downtime. The study details the design and installation of high-efficiency photovoltaic systems equipped with smart charge controllers, which provide a steady, low-current charge suitable for lead-acid batteries in high-temperature, remote oilfield settings. Over 12 months, the methodology involved retrofitting a fleet of cementing bulkers and monitoring performance metrics, including the frequency of battery replacements, costs of spare parts, technician labor hours, and equipment readiness rates. Results showed a complete halt to battery replacements on the retrofitted units in 2024, an 86% reduction in spare part costs, a 33% decrease in maintenance labor hours, and a 99% readiness rate, resulting in a significant decrease in downtime and maintenance costs. The approach offers a scalable and environmentally friendly solution that supports corporate ESG goals, minimizes lead-acid battery waste, and uses renewable energy in harsh field conditions. This work develops a repeatable model for enhancing equipment reliability and operational efficiency across oilfield service fleets using solar-powered maintenance methods.

Introduction

In the oil and gas industry, cementing bulk equipment is crucial for transporting and storing dry cement used in well construction. These units are often placed at remote operational bases for long periods between jobs, where they face extended inactivity. A common operational issue is the deterioration of onboard lead-acid batteries due to self-discharge during periods of inactivity. This deterioration reduces equipment readiness, raises maintenance costs, and causes unplanned downtime that can disrupt vital construction schedules.

Lead-acid batteries naturally lose charge over time through self-discharge even when disconnected from loads ([Battery University, 2024](#)). This self-discharge rate accelerates in the high ambient temperatures typical of Middle Eastern oil fields, where temperatures often exceed 45 °C. Elevated temperatures increase chemical activity within the battery, leading to sulfation—a process in which lead sulfate crystals form on

the plates, thereby reducing capacity and lifespan (Linden & Reddy, 2002). If not addressed, these batteries can reach a deep-discharge state, rendering them unable to start essential equipment without costly fixes, such as jump-starting, towing, or full replacement.

To address this challenge, this study examines the design, installation, and field performance of a solar panel trickle charging system (Figure 1). The proposed system utilizes high-efficiency photovoltaic panels, intelligent charge controllers, and weatherproof electrical components to maintain battery charge during prolonged periods of inactivity, eliminating the need for human intervention or grid power. By providing a steady, low-current charge that matches the battery's self-discharge rate, the system keeps voltage within the safe float range, preventing sulfation and ensuring reliable equipment start-up. The passive nature of this approach is especially well-suited to the remote, off-grid, and high-temperature environments typical of Middle Eastern oil fields, supporting operational reliability while promoting sustainability using renewable energy.

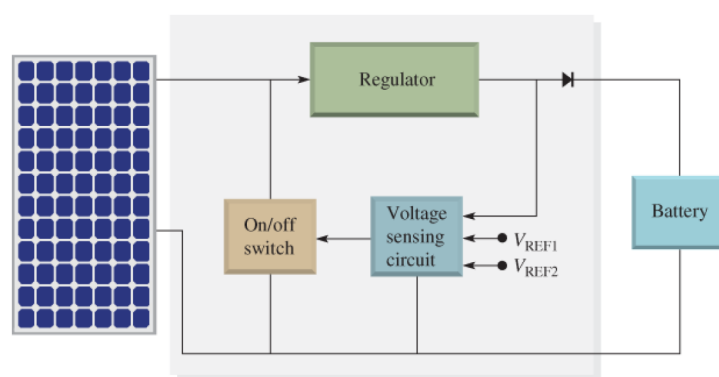


Figure 1—Solar Panel Trickle Charging System.

Literature Review

Battery degradation during storage is a well-documented issue that significantly affects operational readiness. Self-discharge is a chemical process that persists even when no load is applied, with rates doubling at higher temperatures (Battery University, 2024; Linden & Reddy, 2002). In desert oilfield environments, ambient temperatures frequently exceed 45°C, further accelerating battery self-discharge and electrolyte evaporation.

Research into solar-powered battery charging has shown promising results for addressing such problems. Agarwal and Saxena (2018) described the design of intelligent, solar-powered battery chargers capable of maintaining battery voltage within safe ranges without requiring grid infrastructure. Goel and Chauhan (2021) analyzed photovoltaic charging performance under various irradiance levels, emphasizing the importance of effective charge controllers in preventing overcharging.

Arora and Sharma (2013) reviewed battery management systems (BMS), emphasizing their importance in regulating charge and preventing damage. Viswanathan and Kintner-Meyer (2017) highlighted the value of integrating renewable energy for grid-friendly storage solutions. In oilfield contexts, Smith and Jones (2022) emphasized the importance of enhancing equipment readiness in harsh environments through proactive maintenance strategies. Although lithium-ion systems offer higher energy density (Diouf & Pode, 2015), lead-acid batteries remain dominant in heavy equipment because of their cost, availability, and ruggedness.

Recent work on IoT-enabled solar charging for industrial backup batteries further demonstrates the feasibility of remote, autonomous charging solutions that reduce the need for manual maintenance (Hassan & Saeed, 2020). Building on these insights, this study applies solar-powered trickle charging to cementing bulkers, offering a field-tested approach to decreasing downtime and maintenance costs.

Problem Statement

Cementing bulkers face significant operational challenges due to battery self-discharge when left idle for long periods. These specialized transport and storage units often remain unused for weeks or months, especially between drilling campaigns or during equipment rotation. During such times, batteries do not receive auxiliary charging from engine alternators, leading to voltage drops below critical levels.

Several factors contribute to this degradation: harsh environments with ambient temperatures exceeding 45 °C accelerate self-discharge and electrolyte evaporation. Limited staffing at remote sites means routine battery maintenance is deprioritized or impractical, resulting in unmonitored voltage drops that lead to deep discharge states and sulfation. Additionally, logistical constraints hinder technician access for diagnostics and charging.

The result is a widespread maintenance burden: frequent battery replacements, high spare parts costs, labor-intensive emergency repairs, and decreased operational readiness. Battery failures during mobilization cause delays, increase non-productive time (NPT), and can even lead to contractual penalties. These ongoing costs and risks underscore the need for an autonomous, energy-efficient charging solution that can preserve battery health during extended periods of inactivity without requiring human intervention or access to grid power.

Methodology

This project involved retrofitting selected cementing bulkers with solar panel trickle charging systems designed to maintain battery voltage levels during extended periods of inactivity. The objective was to develop a passive, maintenance-free solution that could reliably operate in remote, high-temperature environments with minimal technician involvement and no dependence on grid power or fuel-based generators.

System Design

Each cementing bulker was equipped with a monocrystalline photovoltaic (PV) panel, rated between 20 W and 40 W, selected based on the expected daily solar irradiance and the battery's self-discharge rate. The panels were mounted on the equipment rooftop with orientation optimized for maximum daily solar exposure, considering typical sun angles in Middle Eastern latitudes. The electrical system included smart pulse-width modulation (PWM) charge controllers, designed to regulate voltage and current, ensuring the battery voltage remained within the optimal float range of approximately 13.2–13.5 V. This prevented overcharging, gas, and unnecessary battery degradation. For safety and durability in harsh conditions, inline fuses, automotive-grade UV-resistant wiring, and IP65+ rated weatherproof enclosures were used to protect all critical components from dust, heat, sand ingress, and moisture.

Installation Workflow

The installation process followed a structured workflow. The process began with site surveys to identify optimal mounting locations and assess shading risks from surrounding structures or equipment. Component selection involved specifying PV panel wattage, controller type, and wiring requirements tailored to each unit's battery bank size. Custom-built metal brackets and clamps were fabricated to ensure secure mounting that could withstand wind loads, vibration during transport, and operational rig movements. Electrical integration included routing cables through protective conduits to prevent mechanical wear, installing inline fuses near battery terminals for short-circuit protection, and ensuring all connections were sealed to prevent corrosion. Charge controllers were housed in weatherproof, labelled enclosures with standardized wiring layouts to simplify future inspection and maintenance. Quality checks included continuity testing, fuse verification, and torque checks on all terminal connections.

Commissioning

Following physical installation, a structured commissioning protocol was implemented. Technicians were trained to verify the proper function of the charge controllers by simulating various light conditions and measuring the resulting voltage and current outputs with calibrated multimeters. Baseline battery voltages were recorded before activation of the solar system to establish reference conditions. System behavior was logged at various times of day to confirm that the charging current responded to changes in irradiance and that the controllers correctly entered float or absorption modes. Installation parameters and serial numbers of components were recorded in a central maintenance database to support future traceability and asset tracking.

Operational Environment

The retrofitted units were deployed in Middle Eastern oilfield environments characterized by high ambient temperatures exceeding 45 °C, frequent dust storms, and idle durations ranging from 2 to 6 months. These environmental challenges necessitated a design focus on minimal maintenance, long-term durability, and consistent charging performance even under partial shading or soiling conditions. Components were selected and tested to meet or exceed automotive and industrial specifications for temperature tolerance and ingress protection.

Monitoring Process

Performance was monitored over 12 months. Technicians performed monthly checks, recording battery voltages, maintenance events, the frequency of battery replacements, and the consumption of spare parts, such as terminals and wiring. Control units without solar charging systems were maintained under similar conditions to provide a comparative baseline. Maintenance logs were digitized and analyzed centrally to identify trends and validate system effectiveness.

Data Analysis Approach

Collected operational data were compared against historical averages to quantify improvements. Key performance indicators (KPIs) included battery replacement frequency per unit per year, annual spare parts costs, technician labor hours required for battery maintenance tasks, and the overall equipment readiness rate defined as the percentage of days units remained operational without battery-related faults. Statistical analysis was used to evaluate the significance of observed improvements, supporting evidence-based decisions about fleet-wide deployment.



Figure 2—Wide-angle view of the cementing bulker in a field yard with the solar panel installed. The solar panel (visible on the unit) operates under real-world conditions, including exposure to dust and high heat. The installation does not interfere with the bulker's regular operation, demonstrating the system's adaptability to the field environment.



Figure 3—Shows a close-up of the system's electrical components mounted on the bulkhead. In this image, a weatherproof enclosure houses the charge controller and inline fuse. Heavy-duty cables from the solar panel connect to this control box, which controls the charging current and safeguards the battery from overcharging. This setup is built for durability and safety in outdoor environments.



Figure 4—The panel, located at the top of the unit, is mounted on a metal support structure and positioned to capture maximum sunlight. This arrangement is intended to recharge the battery while the equipment is continuously idle.

Results and Discussions

The solar panel trickle charging system has proven to be a highly effective, cost-efficient, and environmentally sustainable solution for addressing battery self-discharge challenges in cementing bulkers deployed in remote oilfield environments. By maintaining batteries at optimal float voltage levels during extended idle periods, this system reduces the risk of deep discharge and sulfation, two of the most common causes of premature battery failure. As a result, the approach significantly improves operational efficiency, ensuring that equipment remains consistently ready for deployment even after months of inactivity.

Field data collected over a comprehensive 12-month monitoring period provides compelling evidence of the system's effectiveness. Trial units equipped with solar trickle chargers recorded zero battery replacements in 2024, eliminating failures due to idle-time self-discharge entirely. Spare parts costs related to battery maintenance were reduced by 86% due to the decreased wear and tear on terminals, connectors, and wiring harnesses. Labor requirements also declined markedly, with technician time for diagnostics, jump-starts, and replacements reduced by approximately 33%. Critically, equipment readiness improved to 99%, virtually eliminating downtime caused by battery failures and enabling more reliable, on-demand mobilization for oilfield operations.

Beyond the operational and economic advantages, this project aligns strongly with broader sustainability and environmental objectives. By extending battery service life and reducing replacement frequency, the initiative reduces hazardous lead-acid waste generation and lowers carbon emissions associated with battery manufacturing, transportation, and disposal. Using renewable solar energy in place of grid power or fuel-powered generators for maintenance charging further minimizes the operation's overall carbon footprint. This contributes to corporate ESG goals and demonstrates a practical pathway for decarbonizing routine field maintenance activities.

The success of this project presents a replicable, field-tested model for oilfield service providers seeking to enhance fleet reliability and reduce the total cost of ownership in remote, harsh environments where technician access and grid power are limited. Future research and implementation efforts could focus on optimizing system designs for different climatic regions, integrating advanced lithium-based battery chemistries, and deploying IoT-enabled remote monitoring for real-time performance analytics and predictive maintenance. Collaborative industry initiatives could help standardize solar-powered maintenance strategies across various equipment categories, including fracturing pumps, coiled tubing units, and remote monitoring systems, thereby further amplifying cost savings, reducing environmental impact, and supporting a more resilient and sustainable oilfield maintenance ecosystem.

Table 1—Annual battery replacement trend for cementing bulkers from 2022 to 2024, illustrating the impact of the solar trickle charging initiative (2025 value is projected). The number of battery replacements decreased from 70 in 2022 (when no solar panels were installed) to 35 in 2024, after solar panels were added to 27 units in the fleet. This downward trend highlights a significant improvement in battery longevity and a reduction in maintenance costs as more units are equipped with the solar charging system.

Year	2022	2023	2024	2025
No. of Cementing Bulker (Tripod)	72	92	107	111
Solar Panel Installed	0	7	27	0
Battery Consumption Trend	70	50	35	15
Consumption Trend in %	0	-29%	-50%	-79%
Total Consumption Cost	SAR 44,800	SAR 32,000	SAR 22,400	SAR 9,600

Sustainability and Environmental Impact

Beyond operational savings, the project offers tangible environmental benefits. By decreasing the frequency of battery replacements, the initiative reduces the amount of hazardous lead-acid waste generated and prematurely disposed of. This outcome aligns with corporate ESG (Environmental, Social, and Governance) goals of minimizing environmental impact and promoting resource efficiency. The system's dependence on renewable solar energy eliminates the need for fossil-fuel-powered mobile generators or grid electricity, further reducing the carbon footprint associated with maintenance activities. Reductions in technician travel for emergency replacements also help decrease fuel consumption and related emissions. Overall, the solar trickle charging method provides a sustainable maintenance model that enhances equipment readiness while supporting environmental stewardship and operational decarbonization goals (IRENA, 2020).

Challenges and Mitigation

During implementation, several challenges were identified and proactively addressed. Dust and sand buildup on PV panels was managed through regular cleaning schedules coordinated with routine equipment inspections. To reduce shading, units were strategically parked in open yards to maximize daily sunlight exposure. High ambient temperatures risked overcharging batteries during prolonged sunlight, which was mitigated by selecting smart PWM charge controllers with integrated thermal protection and overcharge cutoff features to keep batteries within safe voltage levels. Mounting stability was also a key focus; custom metal brackets and clamps were designed to withstand vibration, wind loads, and transport shocks. Weatherproof enclosures protected charge controllers and wiring from dust and moisture, ensuring long-term system durability. These strategies ensured reliable system performance under the harsh conditions typical of Middle Eastern oilfields.

Conclusions

The solar panel trickle charging system has proven to be a highly effective, cost-efficient, and environmentally sustainable solution for addressing battery self-discharge challenges in cementing bulkers deployed in remote oilfield environments. By maintaining batteries at optimal float voltage levels during extended idle periods, this system reduces the risk of deep discharge and sulfation, two of the most common causes of premature battery failure. As a result, the approach significantly improves operational efficiency, ensuring that equipment remains consistently ready for deployment even after months of inactivity.

Field data collected over a 12-month monitoring period provides compelling evidence of the system's benefits. Trial units equipped with solar trickle chargers recorded zero battery replacements in 2024, eliminating failures caused by idle-time self-discharge. Spare parts costs related to battery maintenance were reduced by 86%, owing to less frequent damage to terminals, connectors, and wiring harnesses. Labor requirements also declined markedly, with technician time for diagnostics, jump-starts, and replacements reduced by approximately 33%. Perhaps most importantly, equipment readiness improved to 99%, virtually eliminating downtime related to dead batteries and supporting reliable, on-demand mobilization for oilfield operations.

Beyond the operational and economic advantages, this project aligns with broader sustainability and environmental goals. By extending battery service life and reducing replacement frequency, the initiative lowers the volume of hazardous lead-acid waste generated. It reduces carbon emissions associated with battery production, transport, and disposal. The use of renewable solar energy eliminates dependence on grid power or fuel-powered generators for maintenance and charging, further minimizing the carbon footprint of operations.

The success of this project establishes a replicable, field-tested model for oilfield service providers seeking to enhance fleet reliability and reduce total cost of ownership in remote and harsh environments. Future research could focus on optimizing system designs for varying geographic conditions, integrating lithium-based battery technologies, and implementing IoT-enabled remote monitoring for real-time performance analytics. Expanding this approach to additional equipment categories, including fracturing pumps, coiled tubing units, and remote-control systems, could yield even greater cost savings and environmental benefits, supporting a more resilient and sustainable oilfield maintenance strategy.

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